Finiteness of the longitudinal integral and subtraction of the $y' = y$ point in the two–Wilson-line term

November 8, 2026

1 The expression and the apparent divergence

We study the term

$$\sigma(k) \;=\; P\int_{k^+/V}^{1} d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)/\xi}\,\mathcal{K}(x,y;x',y')\,\big[U(y')-U(x')\big]^{ce'}\big[U(y)-U(x)\big]^{ce}\,\rho^{e'}(x')\,\rho^{e}(x), \tag{1}$$

with the prefactor $P = \frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{N_c}{k^+}$, the longitudinal cutoff V, and the kernel

$$\mathcal{K} = \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2} - 2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}. \tag{2}$$

Setting the phase to unity, the ξ integral gives $\int_{k^+/V} d\xi\,(1-\xi)/\xi = \ln(V/k^+) + \mathcal{O}(1)$, so (1) appears to be logarithmically divergent as $V\to\infty$, and on the slice $y' = y$ (where the phase *is* unity for all ξ) the integrand genuinely is. The purpose of this note is threefold: (i) to show that for $k \neq 0$ the integral (1) nevertheless has a finite $V\to\infty$ limit; (ii) to clarify in what sense the point $y' = y$ is and is not divergent; and (iii) to construct an explicit subtraction of the $y' = y$ point, with the counterterm in closed form, so that the subtracted integrand is regular at $y' = y$ even at $V = \infty$ — the form suited for numerical evaluation. All statements are verified numerically in Sec. 6.

2 Structure of the kernel

Write $A = x - y$, $B = x' - y'$, $C = y' - y$, $\hat{C} = C/|C|$. The kernel factorizes through the reflection tensor

$$\mathcal{K} = \frac{A_i}{A^2}\,S_{ij}(\hat{C})\,\frac{B_j}{B^2}, \qquad S_{ij}(\hat{n}) = \delta_{ij} - 2\hat{n}_i\hat{n}_j. \tag{3}$$

Three properties of this structure control everything below.

(i) Boundedness at $C\to 0$. The $(y'-y)^2$ in the denominator of the second term of (2) is cancelled by the two explicit factors of $(y'-y)$ in its numerator: $\mathcal{K}$ is a $0/0$ of homogeneity degree zero in C, direction-dependent but bounded. Since S is an orthogonal (reflection) matrix,

$$|\mathcal{K}| \le \frac{1}{|A|\,|B|} \qquad \text{for all } y'. \tag{4}$$

There is no power singularity at $y' = y$; the only divergence anywhere in (1) is the longitudinal one.

(ii) Pure second harmonic. In the angle θ of $\hat{n}$, $S_{11} = -\cos 2\theta$, $S_{12} = S_{21} = -\sin 2\theta$, $S_{22} = \cos 2\theta$: S is a pure $m = \pm 2$ harmonic (equivalently $S_{ij}(\hat{C})/C^2 = \partial_i \partial_j \ln|C|$ away from $C = 0$), so $\oint d\theta\, S_{ij}(\hat{n}) = 0$.

(iii) Exact angular cancellation. The angular average of the *full* kernel over the direction of C vanishes identically, not only as $|C| \to 0$:

Lemma 1. *Let φ be harmonic on the disk $|y' - y| < r_{\max}$. Then for every $r < r_{\max}$ and every fixed vector A,*

$$\oint_0^{2\pi} d\theta\; A_i\, S_{ij}(\hat{C})\, \partial_j \varphi\big(y + r\hat{C}\big) = 0, \qquad \hat{C} = (\cos\theta, \sin\theta).$$

Proof. Use complex notation $z = C_1 + iC_2 = re^{i\theta}$ and represent a vector v by $v_1 + iv_2$. A reflection acts as $S(\hat{n})v = -e^{2i\theta}\bar{v}$. Writing $\varphi = \mathrm{Re}\sum_{m\geq 0} a_m z^m$ (regular inside the disk), the gradient is $h_1 + ih_2 = 2\partial_{\bar{z}}\varphi = \sum_m m\bar{a}_m \bar{z}^{\,m-1}$, hence $Sh = -e^{2i\theta}\overline{(h_1 + ih_2)} = -\sum_m m a_m r^{m-1} e^{i(m+1)\theta}$. Contracting with A gives a sum of terms $\propto e^{i(m+1)\theta}$ with $m \geq 0$, each of which integrates to zero over θ. □

Since $B_j/B^2 = \partial_{y'_j}\big(-\ln|x' - y'|\big)$ and $-\ln|x' - y'|$ is harmonic in y' for $y' \neq x'$, Lemma 1 applied to (3) gives

$$\oint d\theta_{\hat{C}}\, \mathcal{K}\big(y' = y + r\hat{C}\big) = 0 \qquad \text{for all } r < |x' - y|. \tag{5}$$

(The numerics of Sec. 6 confirm this at machine precision at all radii tested.) In particular, the angular average of the second term in (2) equals the first exactly: the second term is precisely the structure needed for the soft region to decouple.

3 Absence of $\ln V$ at fixed $k \neq 0$

Isolate the ξ integral at fixed y', with $a = k \cdot (y' - y)$ and $\varepsilon = k^+/V$:

$$F_\varepsilon(a) = \int_\varepsilon^1 \frac{d\xi}{\xi}\,(1 - \xi)\, e^{-ia/\xi} = \int_1^{1/\varepsilon} dt \left(\frac{1}{t} - \frac{1}{t^2}\right) e^{-iat}, \tag{6}$$

given in closed form in Appendix A. Its two essential properties are

$$F_\varepsilon(0) = \ln\frac{1}{\varepsilon} - 1 + \varepsilon, \qquad \big|F_\varepsilon(a)\big| \;\leq\; \ln\frac{1}{\max(|a|, \varepsilon)} + c_0 \quad (|a| \leq 1), \qquad \big|F_\varepsilon(a)\big| \leq \frac{c_1}{|a|} \quad (|a| \geq 1), \tag{7}$$

with ε-independent constants $c_{0,1}$: the oscillation of the phase sets in at $\xi \sim |a|$ and decoheres the region $\xi < |a|$, so the effective lower limit of the ξ integral is $\max(\varepsilon, |a|)$. The cutoff V therefore survives only where $|k \cdot (y' - y)| \leq k^+/V$.

Now estimate the y' integral near the dangerous set. Note that the phase is frozen not only at the point $y' = y$ but on the whole line $k \cdot (y' - y) = 0$; the frozen region $|k \cdot C| \leq \varepsilon$ is a *strip* of width $2\varepsilon/|k|$ around that line. Within a neighborhood $|C| < \delta$ of $y' = y$, using the uniform bound (4) on the kernel and boundedness of the Wilson-line factors:

(a) *Frozen strip*, $|k \cdot C| \le \varepsilon$: contribution bounded by $\frac{2\varepsilon}{|k|} \cdot 2\delta \cdot M \, \ln \frac{1}{\varepsilon} \sim \frac{\ln V}{V} \xrightarrow{V\to\infty} 0$. The $\ln V$ in the integrand is real, but it sits on a region whose measure shrinks like $1/V$.

(b) *Active region*, $|k\cdot C| > \varepsilon$: here $|F_\varepsilon| \le \ln \frac{1}{|k\cdot C|} + c_0$ *independently of* V, and $\int_{|C|<\delta} d^2C \, \ln \frac{1}{|k\cdot C|} < \infty$ because a logarithm is integrable in one dimension.

Since $\ln(1/\varepsilon) \le \ln(1/|a|)$ exactly where the cutoff bites, $\ln(1/|a|) + c_0$ is a V-independent integrable majorant and dominated convergence applies:

$$\boxed{\lim_{V\to\infty} \sigma(k) \text{ exists and is finite for every fixed } k \neq 0.} \tag{8}$$

No $\ln V$ remains. The would-be rapidity logarithm has been traded for the *transverse* logarithm $\ln \frac{1}{|k||y'-y|}$, integrable over y'. By property (iii) of Sec. 2 even this logarithm is softer than it looks: its coefficient at $y' \to y$ is a pure second harmonic in $\hat{C}$ and drops after the angular part of the y' integration, so the first surviving contribution from $y' \approx y$ comes from the $\mathcal{O}(|C|)$ variation of $U(y')$ and of B_j/B^2.

Remark 1. *The limit is not uniform in k. At $k = 0$ the phase is frozen on the entire plane, the strip argument fails, and the $\ln V$ divergence is genuine, with coefficient $\propto \int_{x,y,x',y'} \mathcal{K}\,[U\cdots][U\cdots]\rho\rho$ (nonzero in general: Eq. (5) kills only the $y' \approx y$ region, not the whole integral, once the y' dependence of the Wilson-line factors is kept). Correspondingly the finite answer at $k \neq 0$ contains $\ln(1/|k| \cdot$ distances$)$: the divergence has been pushed to the corner $k \to 0$, not removed from the theory.*

Remark 2. *The point $y' = x'$, where the first term of (2) behaves as $1/|x' - y'|$, is protected separately: it is multiplied by $[U(y') - U(x')]$, which vanishes linearly there. Likewise $x = y$ is protected by $[U(y) - U(x)]$.*

4 The slice $y' = y$: divergent integrand, convergent integral

On the slice $y' = y$ the phase equals 1 for all ξ and the integrand carries the factor $F_\varepsilon(0) = \ln(V/k^+) - 1 + k^+/V \to \infty$: the *integrand* has no finite $V \to \infty$ limit at that point. This does not make the *integral* divergent: $\{y' = y\}$ (and the line $k \cdot C = 0$) has zero two-dimensional measure, and by the estimates of Sec. 3 the contribution of a neighborhood $|C| < \epsilon$ is bounded by $\epsilon^2 \ln(1/\epsilon) + \text{const} \cdot \epsilon$, *uniformly in* V. As a function of y' the ξ-integrated integrand is a $\ln(V/k^+)$ plateau of width $\sim k^+/(|k|V)$ sitting on the V-independent profile $\ln \frac{1}{|k||y'-y|}$; the plateau contributes $\sim V^{-1} \ln V \to 0$. (Compare $\int_0^1 dy \, \ln \frac{1}{y+\epsilon}$: the integrand diverges at $y = 0$ as $\epsilon \to 0$, the integral does not.) The pointwise limit $V \to \infty$ and the y' integration do not commute at $y' = y$; exchanging them for the *value* of the integral is justified by dominated convergence.

5 Subtraction of the $y' = y$ point

For numerics one wants the integrand itself, not only the integral, to be finite at $y' = y$ at $V = \infty$. Expand the slowly varying factors about $y' = y$, keeping the phase and the angular

tensor exact:

$$\mathcal{K}\left[U(y')-U(x')\right] = \frac{A_i}{A^2}\, S_{ij}(\hat{C})\, \frac{(x'-y)_j}{(x'-y)^2}\left[U(y)-U(x')\right] \;+\; \mathcal{O}(|C|). \tag{9}$$

Define the counterterm from the leading term, damped at large $|C|$ by $\chi(|C|)$ (below, $\chi = e^{-C^2/r_0^2}$):

$$\sigma_{\rm CT} = P\int_{k^+/V}^{1} d\xi\, \frac{1-\xi}{\xi} \int_{x,y,x',y'} e^{-ik\cdot C/\xi}\, \chi(|C|)\, \frac{(x-y)_i}{(x-y)^2}\, S_{ij}(\hat{C})\, \frac{(x'-y)_j}{(x'-y)^2}\left[U(y)-U(x')\right]^{ce'}\left[U(y)-U(x)\right]^{ce}\rho^{e'}\rho^{e}, \tag{10}$$

and $\sigma_{\rm reg} = \sigma - \sigma_{\rm CT}$ *at integrand level*. By (9) the subtracted integrand differs from zero at $y' \to y$ by $\mathcal{O}(|C|)$, so after the ξ integration the subtracted y'-density vanishes like $|C|\ln(1/|C|)$ at $y' \to y$: the pole is removed pointwise and in $\sigma_{\rm reg}$ the cutoff may be dropped under the integral.

The counterterm in closed form. The y' integral in (10) is a pure J_2 Hankel transform. Using $\int_0^{2\pi} d\theta\, e^{i(n\theta - x\cos\theta)} = 2\pi(-i)^n J_n(x)$ with $n=2$,

$$\int d^2C\; e^{-iq\cdot C}\, S_{ij}(\hat{C})\, \chi(|C|) = -2\pi\, S_{ij}(\hat{q})\, T(|q|), \qquad T(q) = \int_0^\infty dC\, C\, J_2(qC)\, \chi(C), \tag{11}$$

with, for the Gaussian and for a sharp cutoff $\chi = \theta(r_0 - |C|)$,

$$T_{\rm gauss}(q) = \frac{2}{q^2}\Big(1 - e^{-q^2r_0^2/4}\Big) - \frac{r_0^2}{2}\, e^{-q^2r_0^2/4}, \qquad T_\theta(q) = \frac{2}{q^2}\big(1 - J_0(qr_0)\big) - \frac{r_0}{q}J_1(qr_0). \tag{12}$$

Both satisfy $T(q) \to 2/q^2$ as $q \to \infty$ and $T(q) = \mathcal{O}(q^2)$ as $q \to 0$. With $q = k/\xi$,

$$\boxed{\sigma_{\rm CT} = -2\pi P\, S_{ij}(\hat{k})\left[\int_0^1 d\xi\, \frac{1-\xi}{\xi}\, T\big(|k|/\xi\big)\right]\Sigma_{ij}}, \qquad \Sigma_{ij} = \int_{x,y,x'} \frac{(x-y)_i\,(x'-y)_j}{(x-y)^2(x'-y)^2}\big[U(y)-U(x')\big]^{ce'}\big[U(y)-U \tag{13}$$

Since $T(|k|/\xi) \to 2\xi^2/k^2$ at $\xi \to 0$, the ξ integrand behaves as $2\xi/k^2$ at the lower end: *the counterterm carries no* $\ln V$ and the cutoff has already been removed in (13). In the distributional scheme $r_0 \to \infty$ the ξ integral collapses to $\int_0^1 d\xi\,(1-\xi)/\xi \cdot 2\xi^2/k^2 = 1/(3k^2)$ and

$$\sigma_{\rm CT}^{r_0\to\infty} = -\frac{2\pi}{3k^2}\, P\, S_{ij}(\hat{k})\, \Sigma_{ij}. \tag{14}$$

Remark 3 (Zero residue)**.** *Had one defined the counterterm with the phase frozen to* 1 *as well (the textbook soft subtraction), it would vanish identically:* $\int d^2C\, \chi\, S_{ij}(\hat{C}) = 0$ *by property (ii) of Sec. 2. This is the subtraction-scheme restatement of Sec. 3: there is no rapidity divergence to subtract from this term at* $k \neq 0$*. Eq.* (10) *is a finite,* V*-independent reorganization whose sole purpose is regularity of the integrand at* $y' = y$*.*

Remark 4 (Scheme dependence)**.** r_0 *is a scheme parameter; its dependence cancels exactly between* $\sigma_{\rm reg}$ *and* $\sigma_{\rm CT}$*. A physical choice is the typical dipole/target size. The frozen factor* $[U(y)-U(x')]$ *in* (10) *no longer vanishes at* $y' = x'$*, but the counterterm kernel* $(x'-y)_j/(x'-y)^2$ *is regular there, so no new singularity is introduced. The* $k \to 0$ *sensitivity of* σ *is now explicit in the* $S_{ij}(\hat{k})/k^2$ *of the counterterm, with a matching* $\ln|k|$ *in* $\sigma_{\rm reg}$*.*

6 Numerical verification

All claims were checked in a toy model that keeps every contested structure and drops spectators: $y = 0$ (so $C = y'$), fixed $A = (1, 0.2)$ and $x' = (-0.6, 0.8)$ placed outside the integration domain $|C| < \frac{1}{2}$ (the $y' = x'$ point being separately protected in the full expression), a smooth stand-in $f(C) = e^{-(C-c_0)^2}$, $c_0 = (0.13, 0.21)$, for the Wilson-line and source factors (nonzero and off-center at $C = 0$ so that no accidental symmetry hides anything), $k = (1, 0)$, $\varepsilon = 1/V$, and the exact F_ε of Appendix A (itself verified against direct quadrature of the ξ integral). The scripts are in `notes/checks/`.

(1) The slice diverges pointwise. $F_{1/V}(0) = \ln V - 1 + 1/V$: 3.6152, 8.2104, 12.8155, 17.4207 at $V = 10^2, 10^4, 10^6, 10^8$ — steps of exactly $\ln 100$.

(2) The integral converges (phase active, $k \neq 0$). $I(V) = \int_{|C|<1/2} d^2C\, \mathcal{K}f\, F_{1/V}(k \cdot C)$:

V	10^2	10^4	10^6	10^8
$I(V)$	$-0.21637 + 0.01615i$	$-0.21664 + 0.01616i$	$-0.21679 + 0.01616i$	$-0.21679 + 0.01616i$

While the integrand at $y' = y$ grew from 3.6 to 17.4, the integral moved in the fourth decimal and froze.

(3) Remove the phase and the divergence is real. With $k = 0$: $I = -0.0079, -0.0179, -0.0280, -0.0380$ at the same V values — linear in $\ln V$ (-0.01006 per factor 10^2). The finiteness in (2) is the phase at work, not smallness of the kernel.

(4) Exact angular cancellation. $\oint d\theta\, \mathcal{K}(r\hat{n})$ evaluates to 10^{-16} at every radius tested ($r = 0.2$ down to 2×10^{-4}), confirming Eq. (5) at all r, as Lemma 1 requires.

(5) The subtraction works. With $r_0 = 0.1$ and the frozen factor $f(0)$: the Fourier identity (11)–(12) was confirmed to eight digits componentwise; the counterterm from the closed form (13), -0.0101750, agrees with its direct two-dimensional evaluation at $V = 10^8$, -0.0101754, whose imaginary part vanishes as predicted; the subtracted integral is flat, $\sigma_{\text{reg}} = -0.20663, -0.20649, -0.20662$, at $V = 10^2, 10^4, 10^6, 10^8$; and $\sigma_{\text{reg}} + \sigma_{\text{CT}} = -0.21679229 + 0.01616024i$ reproduces the original $-0.21679230 + 0.01616024i$ to eight digits. Finally, the y'-density after the ξ integration at $V = 10^{12}$, along $C = (\epsilon, 0)$:

ϵ	unsubtracted	subtracted
10^{-2}	$+2.100 - 1.042i$	$+0.0283 - 0.0140i$
10^{-3}	$+3.667 - 1.075i$	$+0.00174 - 0.00051i$
10^{-4}	$+5.248 - 1.079i$	$+0.00020 - 0.00004i$
10^{-5}	$+6.831 - 1.080i$	$+0.000026 - 0.000004i$

The unsubtracted density climbs like $\ln(1/\epsilon)$ toward the $\ln V$ plateau; the subtracted one vanishes like $\epsilon \ln(1/\epsilon)$. The pole is gone pointwise.

A Closed form of the longitudinal integral

With $t = 1/\xi$, $F_\varepsilon(a) = \int_1^{1/\varepsilon} dt\,(t^{-1} - t^{-2})e^{-iat}$. For $a = 0$, $F_\varepsilon(0) = \ln(1/\varepsilon) - 1 + \varepsilon$. For $a \neq 0$, writing $T = 1/\varepsilon$,

$$\int_1^T \frac{e^{-iat}}{t}\,dt = \mathrm{Ci}(|a|T) - \mathrm{Ci}(|a|) - i\,\mathrm{sgn}(a)\big[\mathrm{Si}(|a|T) - \mathrm{Si}(|a|)\big] \;\equiv\; g_1, \tag{15}$$

$$\int_1^T \frac{e^{-iat}}{t^2}\,dt = e^{-ia} - \frac{e^{-iaT}}{T} - i\,a\,g_1 \;\equiv\; g_2 \qquad \text{(by parts)}, \tag{16}$$

$$F_\varepsilon(a) = g_1 - g_2. \tag{17}$$

The bounds (7) follow from the asymptotics of Si, Ci: the oscillation of e^{-iat} truncates the logarithmic growth of g_1 at $t \sim 1/|a|$, so $F_\varepsilon(a) = \ln\big(\min(1/|a|,\, 1/\varepsilon)\big) + \mathcal{O}(1)$ uniformly, with $1/|a|$ decay for $|a| \gtrsim 1$.